

Muhammad Hamza Azam

AI Engineer · Agentic AI Developer · Python · Automation · Machine Learning

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PROFESSIONAL SUMMARY

AI Engineer and Agentic AI Developer with proven experience building intelligent automation systems, machine learning pipelines, and LLM-powered applications. Holds certifications in Agentic AI and Data Science (both completed June 2026), with 20+ GitHub projects spanning AI agents, n8n workflow automation, FastAPI backends, and end-to-end ML development. Focused on delivering practical, real-world AI solutions that create measurable business impact.

TECHNICAL SKILLS

Programming & Backend: Python (Pandas, NumPy, Scikit-learn, TensorFlow, Keras, FastAPI), C#, SQL, REST APIs

AI & Agent Development: Agentic AI, AI Agents, LLM Workflow Design, Prompt Engineering, LangGraph, LangChain

Automation: n8n, Gmail Automation, Workflow Automation, API Integrations

Machine Learning: Regression, Classification, Clustering, Feature Engineering, Hyperparameter Tuning, Model Evaluation

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Power BI, Streamlit, Excel, Google Sheets

Tools & Platforms: GitHub, VS Code, Jupyter Notebook, Postman

EXPERIENCE

Machine Learning Intern — Remote

September 2025 – November 2025

- Developed and evaluated supervised ML models using Scikit-learn, achieving strong predictive accuracy on real-world datasets.
- Conducted comprehensive exploratory data analysis, uncovering actionable patterns and presenting insights to stakeholders.
- Improved model-ready data quality through advanced preprocessing pipelines, reducing data noise by ~30%.
- Practiced Deep Learning fundamentals using TensorFlow and Keras for classification tasks.

Data Science Intern — Remote

October 2024 – November 2024

- Cleaned and transformed raw datasets, reducing inconsistencies and enabling reliable downstream analysis.
- Built and maintained analytical dashboards in Power BI, reducing manual reporting time for the data team.
- Collaborated with senior data scientists on project documentation and data-driven stakeholder presentations.

Order Placement Specialist — Naheed Supermarket

- Managed 500+ online orders with high accuracy, improving customer experience by 20% and reducing processing errors by 15%.

Crew Trainer — McDonald's

- Trained 15+ employees, boosting team efficiency by 30% and maintaining 90%+ customer satisfaction ratings.

SELECTED PROJECTS

AegisX — AI-Powered Crypto Trading Bot (Python · FastAPI · TradingView · Live Market APIs)

- Built a real-time crypto analysis bot integrating live market APIs to fetch, process, and display price data.
- Implemented TradingView charting library for live visual dashboards served via FastAPI REST endpoints.
- Designed scalable RESTful API architecture enabling responsive, data-driven trading decisions.

Gmail AI Automation Agent (n8n · Gemini API · Workflow Automation)

- Built an automated email workflow using n8n that classifies incoming Gmail messages by intent and priority.
- Integrated the Gemini API to auto-generate context-aware draft replies, reducing manual email handling time.

Mall Customer Segmentation Dashboard (Python · Streamlit · K-Means · DBSCAN · Scikit-learn)

- Applied K-Means and DBSCAN clustering to segment customers by purchase behaviour and spending patterns.
- Deployed an interactive Streamlit web application enabling dynamic cluster exploration and visualisation.

SpaceX Launch Analysis & Prediction (Python · Scikit-learn · Matplotlib · Seaborn)

- Merged and cleaned SpaceX launch and weather datasets; built ML models predicting launch success with high accuracy.
- Visualised launch metrics and feature importance to surface data-driven insights for stakeholders.

Billing & Inventory Management Application (C# · Desktop Application)

- Developed a full-featured desktop application for small business billing, stock management, and automated reporting.

Additional 20+ projects on GitHub: Titanic ML, Stock Market Analysis, Airways Analysis, Student Performance, Employee & Café Management Systems, E-commerce Analysis, and more.

EDUCATION

Bachelor's in Data Science — Sindh Madressatul Islam University (SMIU) (2025 – Present)

Agentic AI Certification — Saylani Mass IT Training Programme (2025 – Jun 2026)

Data Science & AI Certification — Saylani Mass IT Training Programme (2024 – Jun 2026)

Intermediate – Pre-Engineering — Islamia Govt. Science College (2021 – 2022)

CERTIFICATIONS

- Google Data Analytics Professional Certificate — Coursera (Issued: May 2025)
- Agentic AI Certification — Saylani Mass IT Training Programme (Completed: June 2026)
- Data Science & AI Certification — Saylani Mass IT Training Programme (Completed: June 2026)

CORE COMPETENCIES

Problem Solving · Critical Thinking · Team Collaboration · Time Management · Fast Learner · Ownership Mindset

LANGUAGES

Urdu — Native · English — C1, Very Good Command